

Margeaux Corrigan

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EDUCATION

Brown University, Sc. B. Electrical Engineering, 4.0/4.0 GPA

Providence, RI | **Expected Graduation May 2029**

Relevant Courses: Ordinary Differential Equations, Functional Programming, Statics & Dynamics, Data Structures & Algorithms

TECHNICAL SKILLS

Programming: C, Python (NumPy, Pandas, Matplotlib), Ada, MATLAB, Java, OCaml, Racket

Embedded Systems: Bare-metal firmware (RP2040, STM32), register-level programming, I2C/SPI/UART

Hardware: Schematic design(KiCad), soldering, CAD & CAM(SolidWorks, Fusion 360), FEA, 3D printing, sheet metal fabrication

Tools: Git/GitHub, CMake, OpenOCD, CMSIS-DAP, Saleae Logic, oscilloscope, serial monitor, Alire

RESEARCH

Interactive 3D Vision & Learning Lab (IVL), *Paid Researcher*

Providence, RI | June 2026 – Present

Advisor: Srinath Sridhar, Assistant Professor | Brown University

- Supporting multi-camera rig development: PCB schematic & layout design, CAD modeling, & 3D-printed enclosure fabrication

PROFESSIONAL EXPERIENCE

AdaCore, *Technical Mentorship*

Providence, RI | December 2025 – Present

Mentor: Oliver Henley, GNAT Academic Coordinator | AdaCore

- Developed bare-metal Ada firmware for STM32G431 via raw register access from parsed ARM SVD files

Persistence Labs, *Strategy Intern*

Charleston, SC | May 2024 – October 2025

- Facilitated beta testing with high school test groups, collected feedback across 10+ iterations to identify usability issues
- Presented findings to founder/CEO and engineering team, directly informing feature prioritization and UI iteration

MIG3 Inc., *Part-Time Technical Assistant*

Irvine, CA | May 2022 – August 2025

- Provided technical support for office software and built Excel-based data tools, improving internal workflow efficiency

LEADERSHIP & ACTIVITIES

Brown Rocketry Club, *Avionics Team Lead, Co-President*

Providence, RI | September 2025 – Present

- Contributed to Brown Rocketry's first-ever IREC acceptance as sole avionics developer on the competing payload
- Designed IREC avionics electrical architecture and wiring diagrams for power distribution, sensor interfaces, and flight systems
- Performed FEA on avionics bay components and developed CAM-based manufacturing plans in Fusion 360
- Secured industry sponsorship by leading partner outreach and presenting club technical work to external stakeholders

Brown Space Engineering, *Antenna Team Member*

Providence, RI | March 2026 – Present

- Designing communications antennas (Nitinol dipole, copper patch), including specification and integration constraints
- Establishing feed architecture, impedance matching strategy, and EMC constraints on adjacent and connected hardware

AI & Robotics Ethics Society, *Research Team Member*

Providence, RI | September 2025 – Present

- Designing a survey and assessment to evaluate the effectiveness of Brown's Socially Responsible Computing curriculum
- Collaborating with the Computer Science Department and teaching assistants to collect and analyze student feedback

Project Amplify, *Founder & Director*

Irvine, CA | May 2024 – September 2025

- Founded a summer AI literacy program across 2+ school districts, designed and taught curriculum from scratch
- Taught students to build basic image recognition AI, introducing Python coding fundamentals and model evaluation.

ENGINEERING PROJECTS

Bare-Metal Driver Stack, *Independent Project*

RP2040, C | May 2026 – Present

- Developed GPIO, SysTick, UART, I2C, & BMP390 drivers for RP2040 in C using register-level programming without HAL
- Verified UART and I2C via logic analyzer and serial monitor; ISR-driven SysTick timing and atomic GPIO validated
- Designed layered driver structure separating register maps from driver logic for future Cortex-M portability

Rocket Payload Sensor System, *Sole Developer*

RP2040, Ada | February 2026 – April 2026

- Developed bare-metal Ada firmware for an RP2040 rocket payload; work mentored by AdaCore Engineering
- Deployed payload on a live high-power rocket launch; logged repeatable 12+ G flight data, validating system end to end
- Integrated BMP390 and LSM9DS1 sensors over I2C; logged 23+ hours of continuous flight and ground data
- Developed SPI flash logging pipeline for W25Q128 onboard storage; validated write/erase/read-back on hardware
- Debugged I2C addressing, SPI clocking, UART data loss, and power sequencing issues across embedded hardware and firmware

AWARDS & GRANTS

Brown UTRA Research Award, 2026 | **Brown Design Workshop Grant**, 2026 | **Taco Bell Ambition Accelerator Grant**, 2025

PERSONAL INTERESTS

Activities: Ham radio, Girl Scouts(Bronze and Silver Award), reading novels, crochet, cooking, casual bowling, and plate spinning.